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Investigating the utility of 3D-intraoral scan imaging to score plaque compared to visual clinical assessment

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ABSTRACT

Aim: To determine the diagnostic accuracy between intraoral scan-derived plaque assessment (IOS) compared with clinical plaque assessment as the reference standard, before and after plaque disclosure.

Methods: Single-site, cross-sectional study. Eligible participants were adults with a minimum of 20 natural teeth. Following IOS, undisclosed plaque was assessed clinically on all scorable teeth at four sites per tooth using Silness and Löe plaque index. Disclosing agent was applied and plaque scoring repeated, followed by a second IOS. After a wash-out period, the clinical examiner assessed plaque scores on-scan, a second examiner also assessed plaque scores on-scan.

Results: 60 participants aged 19–74 were recruited. Agreement between clinical pre- and post-disclosure scores was poor, with a large increase in plaque detected following use of disclosing agent. Compared to clinical assessment, IOS plaque assessment demonstrated high sensitivity pre- and post-disclosure (90.4, 96.1% and 91.0, 93.9% respectively), while specificity was lower, particularly for undisclosed plaque (30.3, 44.9%), which improved post-disclosure (49.2, 55.9%). When pre-disclosure scores were compared to post-disclosure scores on-scan, sensitivity remained high (80.8, 86.4%), while specificity was lower (38.4, 63.8%). Intra-examiner agreement was higher than inter-examiner agreement, with both increasing following plaque disclosure: intra-examiner agreement increased from 87.5% to 91.5%, and inter-examiner agreement increased from 79.7% to 85.7%. Overall, IOS-derived plaque assessment identified a greater proportion of plaque-positive sites than clinical assessment, with the degree of overestimation reducing following plaque disclosure (29.4% pre disclosure; 14.1% post-disclosure).

Conclusion: IOS-derived plaque assessment demonstrated high diagnostic sensitivity for detecting plaque both pre- and post-disclosure. However, lower specificity indicates a tendency to overestimate the presence of plaque, although diagnostic performance improved following plaque disclosure.

Clinical significance: IOS-derived plaque assessment demonstrates good diagnostic capability for plaque detection and offers potential clinical advantages over conventional visual assessment by enabling the capture and review of plaque assessment at subsequent visits. IOS may therefore represent a promising tool for plaque assessment in clinical dental practice.

1. Introduction

Periodontal diseases are a group of conditions that affect the supporting tissues of the teeth [1]. Gingivitis is the first manifestation of gingival inflammation induced by plaque accumulation and it is a necessary precursor to periodontitis in susceptible individuals [2,3]. Periodontitis is an irreversible, chronic inflammatory condition that results in the loss of the supporting bone and connective tissue. If left untreated, periodontitis may result in tooth loss and it is also associated with an increased risk of all-cause mortality [4]. The primary aim of treatment is therefore the prevention of periodontitis, which can be achieved by eliminating gingivitis through self-performed oral hygiene measures and professional mechanical plaque removal [5].

Gingival inflammation is determined clinically by the presence of

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marginal bleeding, which occurs as bleeding from the superficial gingival tissue rather than from the apical portion of the pocket [6]. Its presence typically reflects a response to plaque accumulation at the gingival margin and it is therefore an indicator of the effectiveness of self-performed plaque control. Consequently, the initial phase of periodontal therapy should focus on controlling gingival inflammation by supporting patients to improve their oral hygiene [7].

Recording plaque and bleeding levels is important for patient monitoring and to help patients achieve and maintain optimal oral hygiene [8]. However, the detection of plaque can be challenging, as plaque is generally colourless. Disclosing agents can be used to stain plaque and visualise deposits. They are recommended as a way to help patients identify areas of plaque accumulation and their use is recommended as an aid in delivering oral hygiene advice [9]. Disclosed plaque assessment is widely used as the comparative reference standard in research evaluating digital plaque detection methods [10], particularly for evaluating agreement between clinical and IOS-derived plaque assessment [11–14]. In clinical practice, dental professionals typically detect plaque by visual examination and the use of a dental explorer alone, without the aid of a disclosing agent, as their application is perceived as time-consuming and cumbersome [15]. However, this may represent a missed opportunity to maximise plaque reduction and disease stability when combined with oral hygiene instruction [16]. Consequently, there is a need for a reliable and convenient method of recording plaque that can be integrated into everyday clinical practice.

Clinical assessment of plaque also requires the use of a plaque index. Numerous indices have been developed for use clinically and in research [17]. Commonly, practitioners use a binary index, scoring the presence or absence of plaque at four sites per tooth [18], which captures sufficient detail for clinical practice oral hygiene advice. More detailed indices have been developed, particularly for research purposes, such as the Turesky modified Quigley-Hein plaque index [19] and Rustogi modified Navy plaque index [20]. They provide more information than a binary index, as they quantify the distribution of plaque across the tooth surface and weight scoring toward the gingival margin due to the importance of this area in the initiation of gingivitis [17]. However, all plaque indices require an observer to assess the presence or absence of plaque and are therefore inherently subjective, depending on examiner reliability [12,21].

Computerised methods have been shown to have the potential to overcome current challenges and offer a more reliable and objective means of determining clinical plaque levels [10]. Two-dimensional intraoral photography has demonstrated utility for image based plaque assessment [10]. However, plaque assessment using 2D-images may be affected by image standardisation and uneven illumination [22], which may make reproducible full-mouth plaque assessment more difficult. Within this context, intraoral scanners (IOS) are of particular interest. IOS have a wide range of applications within dentistry and have demonstrated their diagnostic capabilities for dental caries detection, tooth wear monitoring, gingival recession, and can produce images of sufficient quality to accurately diagnose gingivitis [23–25]. The ability of IOS to capture 3D colour images has also been shown to enable objective quantification of plaque, helping to overcome the inherent subjectivity of visual plaque scoring [26].

Previous studies comparing IOS-derived and clinical plaque scoring have shown generally good agreement [12,14], with the recent study by Gkavela et al. [14] showing moderate to strong agreement between the two methods with and without disclosing solution. However, comparison across studies is limited due to methodological differences, including the use of different plaque indices, non-standardised scan viewing protocols and a predominant focus on agreement rather than diagnostic accuracy using sensitivity and specificity, which has not been systematically evaluated for IOS-derived plaque assessment pre- and post-disclosure. In addition, the relatively small sample size of the studies restricts generalisability, highlighting the need for a more robust evaluation in a larger cohort.

The aim of the present study was to determine the diagnostic accuracy of IOS-derived plaque assessment compared with predefined clinical plaque assessment as the reference standards, before and after plaque disclosure, using sensitivity and specificity. Intra- and inter-examiner agreement for IOS-derived plaque assessment were also evaluated. In the present study, clinical plaque assessment was selected as the pragmatic reference standard. Diagnostic accuracy measures should be interpreted as the performance of IOS-derived plaque assessment relative to the predefined clinical reference standard (undisclosed or disclosed), rather than as absolute measures of the true presence or absence of plaque.

2. Methods

2.1. Study overview

This was a single-site, blinded (only to examiners when assessing the intra-oral scans), cross-sectional study using a CE-marked IOS (TRIOS 5, 3Shape TRIOS A/S, Copenhagen, Denmark), conducted at a UK dental hospital. The study was conducted in accordance with ISO 14,155:2020 – Clinical investigation of medical devices for human subjects - Good Clinical Practice (GCP), and ethical principles outlined in the Declaration of Helsinki. Ethical approval was granted by 24/YH/0130: Yorkshire & The Humber – Bradford Leeds Research Ethics Committee. The study was registered on the ISRCTN database (ISRCTN11821756).

2.2. Participants and recruitment

Participants were recruited from a UK University Clinical Trials Unit's database of individuals who had expressed an interest in taking part in dental clinical trials, and individuals who had responded to clinical study poster advertisements. Interested participants were sent a study information sheet and were invited for screening. At the screening appointment, participants gave written informed consent, then the primary study dentist (Examiner 1) assessed eligibility. Inclusion criteria for eligible participants were adults aged ≥18 years, in good general health, a minimum of 20 natural teeth not including crowns, bridges or implants, willing and able to give informed consent. Exclusion criteria included use of any orthodontic appliance, obvious signs of untreated dental disease which in the opinion of the examiner would interfere with the validity of the study, or a known allergy to any ingredient in the disclosing agent.

2.3. Examiners

Two examiners were used in this study. The primary study dentist (Examiner 1) consented participants, conducted a clinical examination to confirm participant eligibility, completed clinical plaque scoring, performed the IOS, and, after an interval of at 2 weeks, completed IOS-derived plaque scoring, followed by re-scoring after of at least 2 weeks. Examiner 2 was blinded to the clinical assessment and only conducted the IOS-derived plaque scoring once. Both examiners received the same training and were calibrated prior to the start of the study. Training included scoring disclosed and undisclosed plaque on live participants and using IOS-derived scans.

2.4. Study procedure

Following screening, an IOS was used to capture scans of both dental arches by Examiner 1, followed by clinical scoring of undisclosed plaque on all scoreable teeth including 3rd molars at four sites per tooth: Mesio-buccal, mid-buccal, disto-buccal and lingual or palatal, using Silness and Löe plaque index (SLPI) scores of 0, 2 and 3 as shown in Table 1 [27]. A score of 1 was not recorded for undisclosed plaque assessment, as the use of a probe would disrupt the plaque prior to the second scan following use of disclosing agent. All tooth surfaces were then disclosed. The Top

Table 1
Silness and Löe plaque index [27].

Score	Description
0	No plaque
1	A film of plaque adhering to the free gingival margin and adjacent area of the tooth. The plaque may be seen in situ only after application of disclosing agent or by using a probe on the tooth surface
2	Moderate accumulation of soft deposits within the gingival pocket, or on the tooth and gingival margin, visible to the naked eye
3	Abundance of soft matter within the gingival pocket and/or on the tooth and gingival margin

Dent Rondell Röd solution; 5% erythrosine dye disclosing agent (Directa AB, Sweden) was used for 15 s then expectorated. Scoring was then repeated by Examiner 1 at the same four sites per tooth using SLPI scores 0, 1, 2 and 3. Following this, a second scan was taken of the disclosed teeth for both arches. Participant involvement ended at this point.

After a wash-out period of two weeks, Examiner 1 completed the on-scan plaque scoring for both undisclosed and disclosed scans. After a second wash-out period of two weeks, Examiner 1 repeated the on-scan plaque scoring. Examiner 2 also performed IOS-derived plaque scoring for undisclosed and disclosed scans. IOS-derived plaque assessment was carried out on all scoreable teeth at the same four sites per tooth using the corresponding SLPI scores for pre- and post-disclosure scans. Pre-disclosure scans were scored before post-disclosure scans to avoid bias.

A schematic overview of the study workflow, including sequence of clinical and IOS-derived plaque assessments, is presented in Fig. 1.

2.5. Statistical methods

Sufficient participants were recruited onto the study in order to meet the proposed minimum of 60 participants. Scoring a minimum of 20 teeth at four sites per tooth results in at least 80 sites scored per participant. Thus a minimum total of 4800 scorable sites was anticipated. As no previous studies reporting the diagnostic accuracy of IOS-derived plaque assessment could be identified, sample size estimation was informed by preliminary analyses using data from Daly et al. [23]. Based on a review of diagnostic performance, acceptable clinical thresholds for both sensitivity and specificity were set at 0.75, corresponding to false-positive and false-negative rates of 25% [28]. The sample size was considered sufficient to provide adequately narrow confidence intervals to be informative.

The primary and secondary endpoints were sensitivity and specificity, calculated with two-sided 95% confidence intervals. Bootstrap intervals were used as the data were analysed at site level, and sites within the same mouth cannot be assumed to be independent. For diagnostic accuracy analyses, clinical plaque assessment was predefined as the reference standard. For analyses of pre-disclosed IOS-derived plaque assessment, the corresponding clinical pre-disclosed plaque assessment was used as the reference standard. Similarly, for analyses of post-disclosed IOS, the corresponding clinical post-disclosed plaque assessment was used as the reference standard. For each diagnostic accuracy analysis, true positives, true negatives, false positives and false negatives were defined according to the relevant reference standard for that comparison. A true positive was defined as a site classified as plaque-positive by both the index test and the relevant reference standard, while a true negative was defined as a site classified as plaque-negative by both. A false positive was defined as a site classified as plaque positive by the index test but plaque-negative by the reference standard. In contrast, a false negative was defined as a site determined as plaque-negative by the index test but plaque-positive by the reference standard.

For the present analyses, intra- and inter-examiner agreement were assessed by dichotomising plaque as visible, using SLPI scores 2 or 3, or not visible, using SLPI scores of 0 or 1. Overall percentage agreement was calculated with 95% confidence intervals for intra- and inter-examiner comparisons [29], In addition, measures of shift within the same examiner and bias between examiners was calculated. Chance-corrected agreement was reported using Scott's Pi [30], also with bootstrap 95% confidence intervals. Missing or indeterminate recorded scores were excluded from the analysis. All analyses were performed using IBM SPSS Statistics Version 27 (IBM Corp., Armonk, NY, USA).

3. Results

61 participants were screened between 15 August and 10 October 2024. In total, 1 failed screening due to an insufficient number of permanent teeth and 60 were recruited onto the study. Age ranged from 19 to 74 years old. The majority of participants were female (70%), most were white (80%), and reported being non-smokers (82%). Full demographic characteristics are shown in Table 2. There we no adverse events.

Clinical plaque scores pre- and post-disclosure are shown in Table 3, at site level (A) and at participant level (B). As anticipated, a large upwards shift in plaque score was observed following the use of disclosing agent. At site level, the percentage of score 0 recorded at all sites decreased from 60.3% pre-disclosure to 22.0% post-disclosure; conversely the percentage of score 3 increased from 3.6% to 16.6%. Correspondingly, 67% of participants had a maximum plaque score of 3 pre-disclosure, increasing to 100% post-disclosure.

Plaque was least prevalent at mid-buccal sites.

An example of clinical and IOS-derived plaque assessment is shown in Fig. 2.

The diagnostic accuracy of IOS-derived plaque assessment for detecting the presence (SLPI score 2 or 3) and absence (SLPI score 0 or 1) of plaque is summarised in Table 4. Sensitivity and specificity of pre-disclosure IOS-derived and clinical plaque assessment are presented using the corresponding post-disclosure assessment, as the reference standard. As expected, clinical scoring pre-disclosure seriously

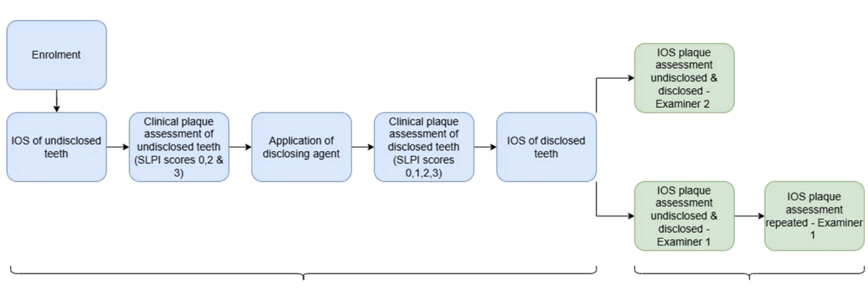

Fig. 1. Schematic overview demonstrating the sequence of clinical and IOS-derived plaque assessments for undisclosed and disclosed plaque, including examiner scoring procedures. Blue-highlighted stages indicate procedures involving participant attendance, while green stages represent IOS-derived plaque assessment independent of participant involvement.

Table 2
Demographic characteristics of study participants.

Characteristic		
Age (years)		
Mean (SD)	37.5 (12.8)	
Range	19.8–74.9	
	Frequency	**Percent**
Gender		
Male	18	30
Female	42	70
Ethnicity		
White	48	80
Mixed	5	8
Asian	2	3
Black	1	2
Other	4	7
Smoking status		
Smoker	7	12
Non-smoker	49	82
Ex-smoker	4	7

underestimated the plaque that is seen post-disclosure, with low sensitivity (58.9%) and high specificity (88.7%). Nevertheless, both examiners achieved high sensitivity but low specificity when scoring pre-disclosure scans relative to what they observed on post-disclosure scans.

For IOS-derived plaque assessment using clinical scores as the reference standard (Table 5), sensitivity was high for both examiners pre-disclosure (Ex1: 90.4%; Ex2: 96.1) and post-disclosure (Ex1: 93.9%; Ex2: 91.0%). However, the specificity was lower, particularly pre-disclosure (Ex1: 44.9%; Ex2: 30.3%) but this improved post-disclosure (Ex1: 55.9%; Ex2: 49.2%). Overall, IOS-derived plaque assessment correctly identified the majority of plaque positive sites, while correctly identifying fewer plaque-negative sites.

The degree of overestimation of plaque identified on-scan by examiner 1 compared with clinical scoring is summarised in Table 6 for each tooth type. Overall, a greater proportion of sites were identified with plaque on-scan than clinically both pre- and post-disclosure. For all tooth types, the degree of overestimation was higher pre-disclosure (29.4%) than post-disclosure (14.1%). Pre-disclosure, overestimation was lowest for incisors and highest for premolars and molars. Post-disclosure, overestimation was lowest for molars and highest for premolars and incisors.

Fig. 2. Clinical photographs A) pre-disclosure B) post-disclosure and IOS-derived images C) pre-disclosure D) post-disclosure.

Table 4
Sensitivity and specificity of pre-disclosure clinical and IOS-derived plaque assessment using post-disclosure scores as the reference standard.

Index → Reference	Sensitivity	95% confidence limits	Specificity	95% confidence limits
Clinical pre → clinical post	58.9%	54.0%, 63.7%	88.7%	86.3%, 91.1%
Ex1 pre → Ex1 post	80.8%	77.9%, 83.7%	63.8%	60.0%, 67.7%
Ex2 pre → Ex2 post	86.4%	83.9%, 88.9%	38.4%	32.1%, 44.7%

Table 5
Sensitivity and specificity of IOS-derived plaque assessment using corresponding clinical plaque assessment as the reference standard.

Index → Reference	Sensitivity	95% confidence limits	Specificity	95% confidence limits
Ex1 pre → clinical pre	90.4%	88.4%, 92.3%	44.9%	41.6%, 48.2%
Ex2 pre → clinical pre	96.1%	94.9%, 97.3%	30.3%	25.8%, 34.9%
Ex1 post → clinical post	93.9%	92.2%, 95.7%	55.9%	50.5%, 61.4%
Ex2 post → clinical post	91.0%	88.1%, 94.0%	49.2%	44.3%, 54.1%

Table 3
Clinical Plaque scores: **(A)** at site level **(B)** at participant level (maximum plaque score).

(A)	**Score**		**Mesio-buccal**		**Mid-buccal**		**Disto-buccal**		**Lingual/palatal**		**Total**	
			Pre	Post	Pre	Post	Pre	Post	Pre	Post	Pre	Post
	0	Count	854	206	1294	674	812	194	766	287	3726	1361
		%	54.7	13.2	83.1	43.3	53.1	12.7	49.9	18.7	60.3	22.0
	1	Count		212		378		207		326		1123
		%		13.6		24.3		13.5		21.3		18.2
	2	Count	644	791	242	379	635	767	712	732	2233	2669
		%	41.3	50.7	15.5	24.3	41.5	50.1	46.4	47.7	36.1	43.2
	3	Count	63	352	21	126	82	362	58	189	224	1029
		%	4.0	22.5	1.3	8.1	5.4	23.7	3.8	12.3	3.6	16.6
	Total	Count	1561	1561	1557	1557	1529	1530	1536	1534	6183	6182
(B)	**Score**		**Mesio-buccal**		**Mid-buccal**		**Disto-buccal**		**Lingual/palatal**		**Total**	
			Pre	Post	Pre	Post	Pre	Post	Pre	Post	Pre	Post
	0	Count	0	0	12	0	0	0	1	1	0	0
		%	0	0	20	0	0	0	2	2	0	0
	1	Count		0		3		0		1		0
		%		0		5		0		2		0
	2	Count	35	4	39	16	33	5	36	11	20	0
		%	58	7	65	27	55	8	60	18	33	0
	3	Count	25	56	9	41	27	55	23	47	40	60
		%	42	93	15	68	45	92	38	78	67	100

Table 6
Percentage of sites with visible plaque (+ve) detected clinically and on-scan by examiner 1, and the degree of overestimation, stratified by tooth type: **(A)** Pre-disclosure **(B)** Post-disclosure.

(A)	**Site**	**Clinical +ve plaque (%)**	**On-scan +ve plaque (%)**	**Overestimation (%)**
	Incisor	37.1%	56.9%	19.8%
	Canine	30.8%	57.3%	26.5%
	Premolar	30.8%	68.9%	38.1%
	Molar	54.9%	88.1%	33.1%
	Total	39.7%	69.1%	29.4%
(B)	**Site**	**Clinical +ve plaque (%)**	**On-scan +ve plaque (%)**	**Overestimation (%)**
	Incisor	50.7%	67.5%	16.8%
	Canine	53.0%	65.8%	12.8%
	Premolar	55.3%	73.5%	18.2%
	Molar	76.5%	84.8%	8.3%
	Total	59.7%	73.9%	14.1%

3.1. Intra- and inter-examiner agreement

Table 7 summarises intra- and inter-examiner agreement in the proportion of sites identified as positive for visible plaque by each examiner using IOS-derived plaque assessment, both pre- and post-disclosure. Overall, the crude percentage agreement and adjusted agreement (Scott's pi) were greater for intra-examiner compared to inter-examiner agreement. Both intra-examiner and inter-examiner agreement improved following plaque disclosure.

Table 8 summarises the proportion of sites identified as positive for visible plaque between the two assessments by examiner 1 and between examiner 1 and examiner 2. There was a small increase of 1.8% in the proportion of sites identified as positive by examiner 1 from the first to second pre-disclosure IOS-derived assessment. The corresponding degree of shift post-disclosure was larger, 3.1%. Pre-disclosure, examiner 2 identified 11.0% more sites as positive for plaque compared to examiner 1. This large inter-examiner bias was substantially reduced post-disclosure to 1.0%.

4. Discussion

This study evaluated the diagnostic accuracy of IOS-derived plaque assessment compared with clinical assessment before and after plaque disclosure. Clinical plaque scoring showed poor agreement between pre- and post-disclosed assessments, with a substantial increase in plaque scores at site level and maximum plaque score at participant level following the use of disclosing agent. This reflects the fact that disclosing agents aid the visualisation of plaque and are normally regarded as the most appropriate clinical reference standard [16].

When compared to clinical scores, on-scan plaque assessments, irrespective of disclosure, were better able to identify true positives than true negatives, with sensitivity consistently above 90%, while specificity was around 50% for disclosed plaque and lower for undisclosed plaque. Based on study predefined diagnostic performance thresholds, sensitivity and specificity values of at least 0.75 (75%) were indicative of acceptable performance [28]. Our low specificity values, reflecting the tendency of on-scan assessment to classify plaque-negative sites as plaque-positive, are consistent with previous studies which have reported higher plaque scores using IOS-derived compared to clinical assessment [11,12].

Intra-examiner was greater than inter-examiner agreement for IOS-derived plaque assessment. For both intra and inter-examiner assessment, the percentage agreement and Scott's pi increased post disclosure, the greatest effect being observed for inter-examiner bias, which was substantially lower on disclosed plaque. This likely reflects that the use of disclosing agent improves visual contrast between plaque deposits and clean tooth surfaces, reducing subjectivity during examiner interpretation of the IOS images.

Previous studies assessing IOS-derived plaque assessment have also reported variability in examiner agreement, although direct comparison is limited due to differences in the agreement statistics used [12,14]. In one of these studies inter-examiner agreement appeared to be influenced by examiner experience and training [14]. Similar inter-examiner variation was also observed when assessing IOS-derived gingival recession measurements compared to clinical measurements using the same statistical methods employed here [25]. These findings suggest that variability in IOS-derived assessment is more likely related to examiner interpretation and clinical judgement rather than limitations of the IOS system and confirm that intra-examiner exceeds inter-examiner consistency, particularly for assessment of visually subjective features such as plaque deposits.

In the current study, clinical visual plaque assessment was used as the reference standard for the diagnostic accuracy analyses. Disclosed plaque is generally considered the most appropriate method for clinical plaque assessment and has been used as the comparative reference standard in most studies that have evaluated digital plaque detection methods [10–14]. However, while the use disclosing agents improve the visibility of plaque, disclosed plaque assessment should not be considered a true gold standard, as plaque detection remains dependent on the plaque index used and examiner subjectivity [31]. Therefore, here undisclosed plaque assessment was also included as a reference standard, accepting that disclosing agent is not universally used in routine dental practice.

In the present study, clinical plaque assessment was based on visual inspection alone and did not incorporate tactile feedback from a probe, as this would affect subsequent IOS assessment. Clinical detection of minimal plaque deposits is more challenging without the use of a disclosing agent due to its colourless nature, thus undisclosed plaque prevalence may have been underestimated. In contrast, IOS-derived plaque assessment benefits from the ability to enlarge, reorient and inspect scan images in detail. Consistent illumination across the scan and the ability to repeatedly inspect the gingival margin from difficult angles likely facilitates the detection of subtle plaque deposits that are harder to identify during clinical assessment, particularly in posterior regions. Therefore, some degree of disagreement between IOS-derived and clinical assessment reflects the differences between the two assessment methods, rather than true misclassification of plaque. This distinction is important when interpreting sensitivity and specificity when clinical assessment is used as the reference standard.

Within the context of the present study, overestimation was used as a comparative measure, referring to how many more sites were identified with plaque in the on-scan assessment as compared with clinical visual assessment, and was quantified to determine whether IOS-derived assessment systematically produces higher plaque scores. Overestimation was observed for IOS-derived plaque assessment both with

Table 7
Intra- and inter-examiner crude and adjusted overall agreement in proportion of sites identified as positive for visible plaque (SLPI scores 2 & 3) by examiners 1 and 2 using IOS-derived plaque assessment, pre- and post-disclosure.

		Crude percentage agreement	95% confidence limits	Adjusted agreement (Scott's pi)	95% confidence limits
Intra-examiner	Pre-disclosure	87.5%	86.1% to 88.8%	0.701	0.672 to 0.730
	Post-disclosure	91.5%	90.3% to 92.8%	0.771	0.745 to 0.798
Inter-examiner	Pre-disclosure	79.7%	77.5% to 82.0%	0.464	0.413 to 0.515
	Post-disclosure	85.7%	83.8% to 87.6%	0.625	0.593 to 0.657

Table 8
Intra-examiner shift and inter-examiner bias in proportion of sites identified as positive for visible plaque (SLPI scores 2 & 3) by examiners 1 and 2 using IOS-derived plaque assessment, pre- and post-disclosure.

		Proportion of sites positive for visible plaque			95% confidence limits
		Ex1b	Ex1a	Shift	
Intra-examiner	Pre-disclosure	70.9%	69.1%	+1.8%	−0.2% to +3.8%
	Post-disclosure	77.0%	73.9%	+3.1%	+1.4% to +4.8%
		Proportion of sites positive for visible plaque			**95% confidence limits**
		Examiner 2	**Examiner 1**	**Bias**	
Inter-examiner	Pre-disclosure	80.1%	69.1%	+11.0%	+8.2% to +13.8%
	Post-disclosure	74.8%	73.9%	+1.0%	−1.1% to +3.0%

[1]Ex1a refers to examiner 1 first IOS plaque assessment, [2]Ex1b refers to the examiner 1 repeat assessment.

and without the use of disclosing agent, indicating a systematic difference relative to clinical assessment rather than random variation. The degree of overestimation was greater for undisclosed IOS-derived plaque assessment, suggesting that visual contrast is important for examiner interpretation when scoring plaque on-scan. Overall, the degree of overestimation decreased for all tooth types following the use of disclosing agent. Interestingly, there were differences across tooth types in the pattern of overestimation pre- and post-disclosure. Pre-disclosure, on-scan overestimation was highest for premolars and molars, whereas post-disclosure, overestimation was highest for premolars but lowest for molars. In a study that only reported post-disclosure findings, on-scan overestimation was highest for posterior teeth [13], findings that contrast with our post-disclosure pattern. Our findings indicate that IOS-derived plaque assessment is influenced by the use of disclosing agent and tooth-specific factors. In addition, scanning noise error can produce optical artefacts such as areas of apparent surface roughness on-scan, and may occur in the presence of inadequate scanning conditions, such as saliva pooling and shadowing and contribute to overestimation [32].

The clinical relevance of plaque overestimation should be considered within the broader context of periodontal assessment and patient management. While IOS-derived assessment identified a greater proportion of plaque-positive sites than clinical assessment, small differences in plaque scores may not necessarily alter clinical decision-making. In practice, periodontal management relies upon the overall distribution of plaque, the presence of gingival inflammation and risk factors rather than isolated plaque-positive sites. Consequently modest overestimation of plaque from IOS-derived plaque assessment may have limited impact on oral hygiene instruction and patient management.

A key strength of this study was the relatively large sample size and inclusion of participants with a wide range of plaque scores. This increases the generalisability of the findings and builds on previous smaller studies [14]. However, several limitations must be considered. In this study, to avoid disrupting plaque deposits examiners were not permitted to use the 3-in-1 air syringe to dry teeth and gingivae, thus saliva accumulated along the gingival margin prior to scanning, hindering clear visualisation of the tooth surface on-scan. In addition, both calculus and plaque stain with disclosing agent, making differentiation difficult and potentially increasing post-disclosure plaque scores on-scan. Calculus is a plaque-retentive factor that is conducive to biofilm rather than a direct initiator of gingivitis itself and should not be regarded as plaque [33,34].

IOS-derived plaque assessment has potential clinical use beyond assessment of a patient's plaque status. Previous studies have reported improvements in gingival health following oral health advice that utilised IOS-based images, and improvements in periodontal parameters following AI-assisted personalised oral hygiene instruction delivered through a smartphone application [35,36]. The high sensitivity observed in the present study suggests that IOS-derived plaque assessment may be clinically useful for identifying plaque deposits to support oral hygiene instruction as part of periodontal care. Although specificity was lower, the clinical value of IOS-derived plaque assessment in this context may depend less on the identification of plaque-free sites and more in identifying plaque deposits that can be used to guide patient-specific oral hygiene instruction.

Scoring plaque from IOS-derived scans is limited by the use of clinical indices, which are designed to interpret plaque seen clinically, not as seen on a scan and rely on the subjective assessment of plaque by the examiner, subjectivity that is amplified for small amounts of plaque [31]. Objective detection methods and automated techniques using artificial intelligence (AI) could overcome these limitations with accurate spatial localization of plaque, therefore avoiding examiner subjectivity [17]. Further studies should compare traditional methods using indices to score disclosed plaque on-scan with AI-driven plaque detection software.

Overall this study demonstrates that the IOS used has good diagnostic capability to detect plaque, characterised by high sensitivity, with a tendency to identify a greater number of plaque-positive sites relative to clinical assessment, reflecting lower specificity when clinical visual assessment is used as the reference standard. Future research should explore the use of software to automate plaque scoring to overcome the limitations of applying clinically based indices to IOS-derived assessment. Incorporating these developments may improve the accuracy and clinical relevance of using IOS for plaque assessment.

CRediT authorship contribution statement

Matthew Wright: Writing – original draft, Methodology, Investigation, Formal analysis. **Natasha West:** Methodology, Investigation. **Pia Elisabeth Nørrisgaard:** Writing – review & editing, Supervision. **Robert G Newcombe:** Writing – review & editing, Supervision, Formal analysis. **Maria Davies:** Writing – review & editing, Supervision. **Nicola X West:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

This study was sponsored by 3Shape TRIOS A/S, which provided financial support and access to the equipment and software used in the study. Nicola West has received financial support from the University of Bristol. Given their role as Editorial Board Member, Nicola West had no involvement in the peer review of this article and has no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor. Pia Elisabeth Nørrisgaard is an employee of 3Shape TRIOS A/S. She holds 3Shape TRIOS A/S shares. She holds patents or patent applications related to the technologies investigated in this study. The sponsor had no role in data analysis, interpretation of the results, or the decision to submit the manuscript for publication. All conclusions are those of the authors.

All other authors have nothing to declare.

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